Declaration of AI assistance

European Journal of Combinatorics

Author-prepared disclosure; not an official publisher form or signature.

Manuscript. The sl(2) weight system at c = -3/32 extends to a graph 4-invariant

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Claude (Anthropic) assisted the original mathematical exploration, proof, text, figures, Lean development and checking code. OpenAI Codex assisted this revision's proof and definition audit, elementary faithful-action argument, literature checks, verification records and submission files. These uses include research and writing assistance. The human authors retain responsibility for the final mathematics and submission decisions.

Human oversight. The submitting author stated on 25 October 2026 that the text is a mix of human-written and AI-generated material and that all parts have been checked and understood by a human. This records that author declaration. Human authors retain responsibility for mathematical accuracy, citations and the final version; AI systems are not human coauthors.

Verification scope. The universal extension theorem and its matrix-trace definition are formalized in Lean 4.34.1 with pinned Mathlib. At the unchanged proof-source commit, the complete build and four endpoint axiom reports passed with only propext, Classical.choice and Quot.sound. Python exact checks were rerun locally; Julia, C and C++ passed in the archived full run. The grouping proof for part (a) and the added faithful-action lemma are supplied in the written argument; no new formalization of the abstract Weyl algebra is claimed.

Supporting record. The separate AI_PROVENANCE.md identifies the documented tasks and tools. No unprovided model version, human signature, confidential transcript or journal approval is asserted.

Policy reference. https://www.elsevier.com/about/policies-and-standards/generative-ai-policies-for-journals

Prepared on 25 October 2026. The author reviews this statement and completes any additional live-portal declarations before submission.